DERIVED PUSHFORWARD COMMUTES WITH THE COHERATOR RIGHT ADJOINTS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4757-4766, immediately following the statement at 4745-4755.

Editorial note. The uniqueness of right adjoints is the only exterior principle used; both preservation inputs are same-chapter.

Allowed standard prerequisites (editorial).

- Quasi-coherence of derived pushforward. Lemma lemma-quasi-coherence-direct-image of this chapter, cited at proof lines 4760-4761. Makes Rf_* DQ_X land in D_QCoh(O_Y) so the adjunction on D_QCoh applies.
- Quasi-coherence of derived pullback. Lemma lemma-quasi-coherence-pullback of this chapter, cited at proof lines 4762-4764. Lf* maps D_QCoh(O_Y) into D_QCoh(O_X), the key input for the adjoint-uniqueness step.

Source reference. Original lemma 08FA: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 790–790. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 08F4: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 620–620. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a quasi-compact and quasi-separated morphism of algebraic spaces over S. If the right adjoints DQ_X and DQ_Y of the inclusion functors $D_{QCoh} \to D$ exist for X and Y, then*

$$Rf_* \circ DQ_X = DQ_Y \circ Rf_*$$

Proof. The statement makes sense because Rf_* sends $D_{QCoh}(\mathcal{O}_X)$ into $D_{QCoh}(\mathcal{O}_Y)$ by Lemma 08FA. The statement is true because Lf^* similarly maps $D_{QCoh}(\mathcal{O}_Y)$ into $D_{QCoh}(\mathcal{O}_X)$ (Lemma 08F4) and hence both $Rf_* \circ DQ_X$ and $DQ_Y \circ Rf_*$ are right adjoint to $Lf^* : D_{QCoh}(\mathcal{O}_Y) \to D(\mathcal{O}_X)$. □

Date: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.